WORKSHEET 3.1

Functions Inside Functions: The Chain Rule

Unit 3: Composite, Implicit and Inverse Functions

Key ideas

- **Chain rule**: $\dfrac{d}{dx}\Big[f(g(x))\Big] = f'(g(x)) \cdot g'(x)$: the derivative of the outside, evaluated at the inside, times the derivative of the inside.
- Work from the outside in. With three layers there are three factors.
- $\sin^3 x$ means $(\sin x)^3$: the outside function is the cube.
- With a table, f' is evaluated at $g(a)$, not at a.

Worked examples

Example 1 · A power of a polynomial

Find $\dfrac{d}{dx}\left[(2x^3+1)^5\right]$.

Solution. $5\,(2x^3+1)^4 \cdot 6x^2$.

$30x^2\,(2x^3+1)^4$.

Example 2 · A root of a trigonometric function

Find $\dfrac{d}{dx}\left[\sqrt{3+\cos x}\right]$.

Solution. $\dfrac{1}{2}(3+\cos x)^{-1/2} \cdot (-\sin x)$.

$-\dfrac{\sin x}{2\sqrt{3+\cos x}}$.

Example 3 · From values

Let $h(x) = f(g(x))$, with $g(3) = 2$, $g'(3) = -4$ and $f'(2) = 5$. Find $h'(3)$.

Solution. $h'(3) = f'(g(3))\,g'(3) = f'(2) \cdot (-4)$.

$h'(3) = -20$.

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Practice

CORE

1. Find $\frac{d}{dx}[\sin(5x)]$.

2. Find $\frac{d}{dx}\left[e^{x^2}\right]$.

3. Find $\frac{d}{dx}\left[(x^2+3x)^4\right]$.

4. Find $\frac{d}{dx}[\ln(4x+1)]$.

EXAM LEVEL

5. Find $\frac{d}{dx}\left[\cos^3 x\right]$.

6. Find $\frac{d}{dx}\left[e^{-2x}\sin(3x)\right]$.

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7. Find $\dfrac{d}{dx}\left[\tan(\sqrt{x})\right]$.

8. Let $f(x) = \sqrt{g(x)}$, where $g(4) = 25$ and $g'(4) = -10$. Find $f'(4)$.

Question 9 · Free-response practice

Let $f(x) = (x^2 - 4)^3$.

(a) Find $f'(x)$.

(b) Find all values of x at which the graph of f has a horizontal tangent.

(c) Write an equation of the line tangent to the graph of f at $x = 1$.

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Worked solutions

1. $5\cos(5x)$.
2. $2xe^{x^2}$.
3. $4\left(x^2+3x\right)^3(2x+3)$.
4. $\dfrac{4}{4x+1}$.
5. $3\cos^2 x\cdot(-\sin x) = -3\cos^2 x\sin x$.
6. $-2e^{-2x}\sin(3x)+3e^{-2x}\cos(3x) = e^{-2x}\big(3\cos(3x)-2\sin(3x)\big)$.
7. $\sec^2(\sqrt{x})\cdot\dfrac{1}{2\sqrt{x}}$.
8. $f'(4) = \dfrac{g'(4)}{2\sqrt{g(4)}} = \dfrac{-10}{10} = -1$.
9. (a) $3\left(x^2-4\right)^2\cdot 2x = 6x\left(x^2-4\right)^2$.
   (b) $x=0$, $x=2$ and $x=-2$.
   (c) $f(1)=-27$ and $f'(1)=6(9)=54$: $y+27=54(x-1)$.

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